

QuantPrep — UI/UX Audit & Remediation

Repository [aditya161205/quant-interview-prep](#) • Next.js 16 / React 19 / Tailwind v4 / Supabase • 5 August 2026

Outcome

The running application was audited against the UI/UX rule set, producing **16 findings**. **14 were fixed**; the 2 mobile-view findings were excluded at your instruction, along with the mobile-specific portions of three others. Work was carried out by four subagents partitioned by exclusive file ownership, then independently re-verified in the browser.

Check	Result
Findings fixed	14 of 16 (2 mobile findings excluded by request)
Files changed	14 modified, 1,151 insertions / 430 deletions
New files / dependencies	None
TypeScript	tsc --noEmit clean
Lint	16 problems on the pristine clone, 15 after — 1 pre-existing error fixed, 0 introduced
Production build	Compiles successfully; all 15 static pages generated
Console errors	None
Working tree	14 modified tracked files, 0 untracked; nothing committed

How the audit was run

- Cloned the repository, installed dependencies and confirmed a clean baseline build.
- Ran the dev server and drove the real browser — measuring DOM geometry, network payloads, focus behaviour and colour contrast rather than reading code alone.
- The deployment has no Supabase credentials, so the practice screens render empty. The API routes were *temporarily* stubbed with a synthetic 1,082-row dataset to exercise the real UI, then fully removed — the final tree contains no stub code.
- Contrast figures were obtained by rasterising the actual computed colours to sRGB via canvas and compositing translucent tints over their real backdrops, not by eyeballing swatches.

Findings fixed, with measured before / after

#	Finding	Before	After
4	Practice list downloaded every matching row and paginated in the browser; no caching.	158 KB per visit and per filter change	3.7 KB per page
	Client-side Status filter could not be pushed to the server.	158 KB	5.5 KB — 4.5 KB id set + 1 KB hydration
3	Problem page refetched the whole list to compute Prev/Next, which the detail response already returned.	2 requests (~150 KB)	1 request

#	Finding	Before	After
	Same effect refetched the list on every solve/bookmark toggle (found during the fix, not in the audit).	1 refetch per toggle	0 – local memo
5	Facets endpoint full-scanned the table on every page load to derive ~15 strings that never change.	Full scan, uncached	Grouped aggregates, cached 24 h
—	Profile page still pulled the entire bank to count it (last surviving instance of #4).	156.8 KB	1.9 KB, identical figures
6	Difficulty badges failed WCAG AA in light mode — light branch used a -600 shade where dark correctly used -400.	3.11 / 2.79 / 3.67 : 1	6.90 / 6.51 / 6.84 : 1
7	No live regions anywhere — answer verdicts, hints and every game result were silent to screen readers.	0 live regions	Verdict, hints + 1 per game
8	One empty state for four situations: no match, backend unconfigured, 401, and network failure all read “No problems match your filters.”	1 message, fake “0 problems”	4 distinct states; count shows —
9	“Reveal solution” out-ranked “Check answer” — the give-up path was the largest, filled CTA.	Reveal 190×44 filled	Reveal 159×40 ghost; Check filled
10	Answer field silently deleted keystrokes outside [0-9./-] — “6 flips” became “6”.	Characters vanish, no message	Input preserved; validated on submit
11	Answer format rule lived only in the placeholder, so it vanished on first keystroke.	Placeholder only	Persistent helper text + label
12	Activity heatmap was mouse-only: 186 cells, none focusable, none carrying any text alternative; no axes; zero level invisible.	0 labelled, 0 reachable; 1.12:1	186 labelled, 1 tab stop; 1.53:1
13	Two control idioms for one job (chips vs. native selects); toolbar orphaned the Company select onto its own row; no clear-all.	Ragged wrap, no reset	Unified chips + Clear all
14	Problem title — the most important column — was the most squeezed between 640–900 px.	about 20 px at 640 px	214 px at 640 px
15	Every game intro duplicated the page header , pushing settings and the Start button below the fold on all six games.	Doc 1212 px; Start ends 1026 px	Doc 952 px; Start ends 774 px
16	Pagination was prev/next only — 44 pages of clicking.	Prev / Next	First, numbers, ellipsis, Last
—	No skip link; ~7 tab stops before content on every page.	None	Skip to content, first tab stop
—	Search input had no accessible name; stale “four games” copy (there are six).	Placeholder only; “four”	aria-label; “six”

A bug found while fixing, not in the original audit

The answer checker treated **a failed request as a wrong answer**. Any non-OK response from `/api/problems/[id]/check` fell through to correct `=== undefined` and rendered “*Not quite — try again or reveal the solution.*”, while a thrown network error silently reset the form to idle.

In practice that means a user whose session expired, or who was briefly offline, is told their correct answer is wrong — and sends them rewriting a correct solution. This is the same error/success conflation as finding 8, sitting on the application's single most important interaction. It now reports the failure honestly and distinguishes it from a wrong answer.

Deliberately excluded (mobile view)

Per your instruction, no change was made to behaviour below the 640 px breakpoint. Each agent was explicitly constrained not to touch the base grid or add mobile navigation. These remain open:

#	Open finding	Evidence
1	No mobile navigation at all. Nav is hidden below 640 px with nothing replacing it, so Practice, Market Making and Profile are unreachable on a phone except by typing URLs.	All 4 nav links measure width 0 at 375 px; the only header button is the theme toggle
2	Floating stopwatch overlaps the primary CTA on the problem page; the page reserves no bottom padding for it.	At 375×812 the pill occupies x155–355 / y750–792 and Reveal occupies x41–231 / y766–810
12a	Heatmap cells shrink to roughly 7 px at 375 px wide.	Axis labels are hidden below 640 px so small-screen rendering is unchanged
13a	Filter block occupies about 1,050 px of height before the first problem row on a phone.	Desktop toolbar was unified; mobile stack untouched
14a	Titles truncate to roughly 14 characters at 375 px.	Only the sm/lg column allocation was rebalanced

Independent verification

Every fix below was confirmed in the browser after the agents reported, rather than accepted on their say-so. Two agent claims were re-derived from scratch; one apparent regression was traced to a measurement artefact (the browser pane had collapsed to `innerWidth: 0`, making text wrap one character per line) and was *not* a real defect.

Area	What was checked, and what was observed
Pagination	?page=22 deep-links correctly; row numbering starts at 526; 25 rows served; controls render First / Prev / 1 / ... / 21 / 22 / 23 / ... / 44 / Next / Last with <code>aria-current="page"</code> ; only that page is fetched.
Status filter	The 5 seeded solved problems were found on pages 1, 4, 16, 36 and 44 — proving the whole set is searched, not just page one. Count reported exactly 5.
Profile	Dials 2/361, 1/361, 2/360 and all five category rows are byte-identical to the full-download version, at 1.9 KB instead of 156.8 KB. A bookmark on page 24 still appears.
Contrast	Re-measured by canvas rasterisation: light mode 6.90 / 6.51 / 6.84 : 1 (all pass AA); dark mode unregressed at 8.84 / 9.82 / 6.33 : 1.
Validation	“6 flips” survives typing intact; submitting it sets <code>aria-invalid</code> , announces a message and makes zero network calls.
Failed check	A 503 from the check endpoint now reports “Couldn’t reach the server — your answer wasn’t checked” instead of “Not quite”.
Skip link	First focusable element; on focus becomes 137×38 at (16,16), position fixed, z-index 60 — above the sticky navbar. Target <code>#main-content</code> exists.

Area	What was checked, and what was observed
Game intros	Card game document height 1212 -> 952 px with Start ending at 774 px; tournament 900 px / 722 px. One H1 per page where there had been a duplicate H2.
Empty state	With credentials absent the list reads “Problem library unavailable — the list is not empty, it is unreachable” and the count shows an em dash rather than a fabricated 0.

Judgment calls worth recording

Two agent recommendations overrode the original audit — correctly

The audit proposed a -700 text shade for light-mode badges. The agent measured amber-700 at 4.46:1 on a hovered row — still failing — and moved all three to -800 for a uniform 6.1:1+. Separately, raising the heatmap zero level alone would have made “no activity” *more* prominent than “some activity” in dark mode (1.76:1 against 1.42:1), so the accent steps were moved as well.

A hydration bug introduced and caught during the work

Putting dates into `aria-label` server-rendered them for the first time, and `toLocaleDateString(undefined, ...)` produced en-GB on the server against en-US on the client. The locale is now pinned. The same trap applies to any component that begins server-rendering a formatted date.

Scope decision on the profile page

The API was kept backwards-compatible so the profile kept working, which left it downloading the whole table — finding 4 surviving in its last consumer. Rather than leave it, the per-topic and per-difficulty counts that the facets route already computed and discarded were exposed, so the page now fetches tallies plus only the problems the user has actually touched. The first attempt introduced three lint problems; those were fixed rather than left behind.

Accessibility approach for the heatmap

186 individually focusable cells would bury the rest of the page behind ~190 tab stops, and a native `title` is not keyboard-reachable. The grid instead uses roving `tabindex` — one tab stop, arrow keys to walk days, Home/End to jump — with the tooltip firing on focus as well as hover, plus a plain-text summary as the non-mouse alternative.

One game was deliberately left silent

The Arrow Game receives no per-round announcement. Its rounds are feedback-free by design, so announcing outcomes would both leak information sighted players do not get and fire roughly 80 interruptions per session. Only the end-of-game summary is announced.

Files changed

File	Change
<code>src/app/api/problems/route.ts</code>	Server-side paging with exact count; id-projection and id-hydration modes; <code>not_configured</code> marker
<code>src/app/api/problems/facets/route.ts</code>	Grouped aggregates replacing the full scan; 24 h cache; per-topic / per-difficulty counts + total

File	Change
src/components/problem-browser.tsx	Server-driven paging, numbered controls, four empty/error states, unified toolbar, clear-all
src/components/problem-detail.tsx	Server prev/next, live regions, submit-time validation, failed-request state, button hierarchy
src/components/profile-view.tsx	Facet-driven totals, targeted id fetch, accessible heatmap with axes, empty-state guidance
src/components/difficulty-badge.tsx	Light-mode text shade raised to clear WCAG AA
src/app/layout.tsx	Skip-to-content link; main landmark given an id
src/app/page.tsx	Stale "four games" copy corrected to six, in two places
6 × *-game.tsx	Intro de-duplication and one scoped live region per game (arrow game: summary only)

Nothing has been committed. The working tree holds 14 modified tracked files and no untracked files; `git diff` is available for review.